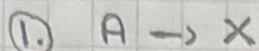


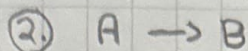
P8-4A.



$r_{1X} = K_1 C_A^{1/2}$

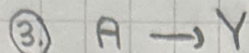
$K_1 = 0.004$

* Assume reactor is PFR



$r_{2B} = K_2 C_A$

$K_2 = 0.3$



$r_{3Y} = K_3 C_A^2$

$K_3 = 0.25$

$$a. S_{B/X} = \frac{r_B}{r_X} = \frac{K_2 C_A}{K_1 C_A^{1/2}}$$

$$C_A = C_{T0} \frac{F_A}{F_T} \frac{P}{P_0} \frac{T_0}{T}$$

$$S_{B/Y} = \frac{r_B}{r_Y} = \frac{K_2 C_A}{K_3 C_A^2}$$

* Assume isothermal and no pressure drop

$$S_{B/XY} = \frac{r_B}{r_X + r_Y} = \frac{K_2 C_A}{K_1 C_A^{1/2} + K_3 C_A^2}$$

$$L \rightarrow C_{T0} = \frac{P}{RT} = 4 \text{ atm}$$

$$0.08206 \frac{\text{atm} \cdot \text{L}}{\text{K} \cdot \text{mol}} \times 300 \text{ K}$$

$$\frac{dF_A}{dV} = r_A = -(r_{1X} + r_{2B} + r_{3Y})$$

$$L \rightarrow r_{1A} = -r_{1X}$$

$$L \rightarrow r_{2A} = -r_{2B}$$

$$L \rightarrow r_{3A} = -r_{3Y}$$

$$C_{T0} = 0.162 \frac{\text{mol}}{\text{dm}^3}$$

$$C_{T0} = C_{A0}$$

$$\frac{dF_B}{dV} = r_B \quad \frac{dF_X}{dV} = r_X \quad \frac{dF_Y}{dV} = r_Y$$

$$\frac{d(S_{B/XY})}{dC_A} = \frac{0.5 K_2 K_1 C_A^{1/2} - K_2 K_3 C_A^{3/2}}{(K_1 C_A^{1/2} + K_3 C_A^2)^2}$$

* Used Mathematica

$$C_A^* = \left(\frac{K_1}{2K_3} \right)^{2/3} = 0.04 \frac{\text{mol}}{\text{dm}^3}$$

This is the C_A used in a CSTR to maximize $S_{B/XY}$.

$$C_i = C_{T0} \frac{F_i}{F_T} \frac{P}{P_0} \frac{T_0}{T}$$

↳ for A, B, X, and Y

$$\hookrightarrow C_{T0} = 0.162 \frac{\text{mol}}{\text{dm}^3}$$

$$\hookrightarrow F_T = F_A + F_B + F_X + F_Y$$

$$\textcircled{d.} \quad X_A = \frac{C_{A0} - C_A^*}{C_{A0}} = 0.753$$

↳ Using a CSTR since the reactor can operate at a constant C_A . (C_A^* will be used)

$$V = \frac{v_0 (C_{A0} - C_A^*)}{-r_A^*} = \frac{10 \text{ dm}^3/\text{min} \cdot 0.162 \frac{\text{mol}}{\text{dm}^3}}{-\left(\underbrace{K_1 C_A^{*1/2}}_{0.004} + \underbrace{K_2 C_A^*}_{0.3} + \underbrace{K_3 C_A^{*2}}_{0.25} \right)}$$

$$= 92.4 \text{ dm}^3$$

$$\textcircled{c.} \quad V = \frac{v_0 C_B}{-r_B} = \frac{v_0 C_B}{K_2 C_A} \Rightarrow C_B^* = \frac{V^* K_2 C_A^*}{v_0}$$

Because first reactor is operated at C_A^*

$$= \frac{92.4 \text{ dm}^3 \times 0.3 \frac{1}{\text{min}} \times 0.04 \frac{\text{mol}}{\text{dm}^3}}{10 \text{ dm}^3/\text{min}}$$

$$= 0.11 \frac{\text{mol}}{\text{dm}^3}$$

$$\cdot C_X^* = \frac{V^* K_1 C_A^{*1/2}}{v_0} = 0.007 \frac{\text{mol}}{\text{dm}^3}$$

$$\cdot C_Y^* = \frac{V^* K_3 C_A^{*2}}{v_0} = 0.004 \frac{\text{mol}}{\text{dm}^3}$$

- e) To achieve 99% conversion a PFR should follow the first CSTR

$$\frac{dX}{dV} = \frac{-r_A}{F_{A0}} \Rightarrow V_{\text{PFR}} = C_{A0} v_0 \int_{0.753}^{0.99} \frac{dX}{K_1 C_A^{1/2} + K_2 C_A + K_3 C_A^2}$$

$$\hookrightarrow C_A = C_{A0}(1-X)$$

$$V_{\text{PFR}} = 94.70 \text{ dm}^3$$

* I used Mathematica to solve the integral.

$$f) \cdot V = \frac{F_{B0} - F_B}{-r_B} = \frac{C_B v_0}{K_2 C_A} \Rightarrow C_B = \tau K_2 C_A$$

$$\hookrightarrow K_2(T) = K_2(300K) \exp \left[\frac{E}{R} \left(\frac{1}{300K} - \frac{1}{T} \right) \right]$$

* C_B can be plotted as a function of temp. I would recommend whatever temp gives a maximum value for C_B .

- g) Since $P_A = C_{ART}$, I would choose $P_A^* (= C_{ART}^*)$ because this would at least give high S_B/X_V which is a good starting point.

P8-10B. *Used Matlab with the help of Claude (AI) to help with syntax and labeling.